SUPPLEMENTARY MATERIAL FOR "WHERE VIOLENCE PERSISTS: LATENT SPATIAL FIELDS FOR THE RESIDUAL RISK OF ARMED CONFLICT AND SOCIAL UNREST"

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This supplement contains the detailed tables and figures referred to in the main paper. Equation, section and table numbers without the prefix S refer to the main paper.

S1. Methods.

S2. Simulation studies.

S3. Application.

TABLE S1

Latent blocks of the latent Gaussian model of Section 3.1 of the main paper, their precision matrices and hyperparameters.

Block	Dimension	Precision	Hyperparameters
fixed effects $\boldsymbol{\beta}, \delta, \boldsymbol{\gamma}, \boldsymbol{\alpha}$	p	$10^{-4}\boldsymbol{I}$	–
country effects $\boldsymbol{u}$	n_c	$\sigma_u^{-2}\boldsymbol{I}$	σ_u
spatial field $\boldsymbol{w}$	n	$\boldsymbol{Q}_s(\rho_w, \sigma_w)$, SPDE precision	ρ_w, σ_w
cell effects $\boldsymbol{e}$	n	$\sigma_e^{-2}\boldsymbol{I}$	σ_e
SVC fields $\boldsymbol{\eta}_k$	n each	$\boldsymbol{Q}_s(\rho_k, \sigma_k)$	ρ_k, σ_k
space-time field $\boldsymbol{v}$	nT_v	$\boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$	ρ_v, σ_v, ϕ

TABLE S2

Boundary treatments on a 60×30 *lattice of* 0.5° *cells with a sea strait of 2, 4 or 6 cells (about 110, 220, 330 km): marginal variance of a coastal and of an inland land cell (nominal 1) and correlation between the two coastal cells facing each other across the strait, from 4,000 exact draws with* $\rho = 400$ *km. "Land only" removes the strait from the lattice; "sea = land" keeps it as ordinary cells without data.*

Strait		Land only	Sea = land	$f = 0.1$	$f = 0.2$	$f = 0.5$
2 cells	var. coast	1.98	1.07	1.80	1.67	1.29
	var. inland	1.10	1.06	1.09	1.11	1.09
	corr. across	-0.01	0.49	-0.03	0.01	0.16
4 cells	var. coast	2.00	1.08	1.77	1.69	1.29
	var. inland	1.06	1.05	1.09	1.09	1.09
	corr. across	0.00	0.28	0.01	0.01	0.04
6 cells	var. coast	2.02	1.04	1.76	1.66	1.26
	var. inland	1.06	1.05	1.08	1.07	1.07
	corr. across	-0.02	0.14	0.01	0.01	0.01

TABLE S3

S1: residual diagnostics and recovery of the field (mean and SD over 20 replicates).

Quantity	Mean	SD
Moran's I of cell residuals, XGBoost as in the original	0.360	0.050
Moran's I of cell residuals, cross-fitted XGBoost	0.637	0.067
$\mathrm{cor}(\hat{w}, w)$, hybrid with $u + w$	0.849	0.044
$\hat{\rho}_w$ (km) [truth 400]	409	50
$\hat{\sigma}_w$ [truth 1]	0.829	0.069

TABLE S4

S2: estimation of the effect α of a national indicator (30 replicates per scenario). Coverage of nominal 95% intervals, and the proportion of replicates in which the interval excludes zero. †Least-squares slope of SHAP values on Z_0, with the 95% block-bootstrap percentile interval and interquartile range.

Estimator	Mean	Emp. SD	Mean SE	Coverage	Reject $\alpha = 0$
No confounding, $\alpha = 0$					
Logistic regression	-0.007	0.107	0.016	0.23	0.77
+ country effects u	0.016	0.109	0.123	0.97	0.03
+ spatial field w	0.002	0.065	0.052	0.93	0.07
+ u + w	0.000	0.063	0.081	1.00	0.00
spatial+ with u + w	0.002	0.072	0.082	0.97	0.03
XGBoost SHAP slope†	-0.006	0.063	0.044	0.80	0.20
Confounding, $\alpha = 0$					
Logistic regression	0.465	0.115	0.018	0.00	1.00
+ country effects u	0.493	0.098	0.078	0.00	1.00
+ spatial field w	0.200	0.098	0.056	0.30	0.70
+ u + w	0.200	0.095	0.077	0.33	0.67
spatial+ with u + w	0.153	0.092	0.079	0.60	0.40
XGBoost SHAP slope†	0.332	0.132	0.056	0.00	1.00
Confounding, $\alpha = -0.3$					
Logistic regression	0.219	0.114	0.017	0.00	1.00
+ country effects u	0.235	0.104	0.079	0.00	0.80
+ spatial field w	-0.090	0.100	0.056	0.20	0.40
+ u + w	-0.091	0.098	0.077	0.37	0.33
spatial+ with u + w	-0.130	0.095	0.078	0.50	0.40
XGBoost SHAP slope†	0.138	0.091	0.049	0.00	0.87

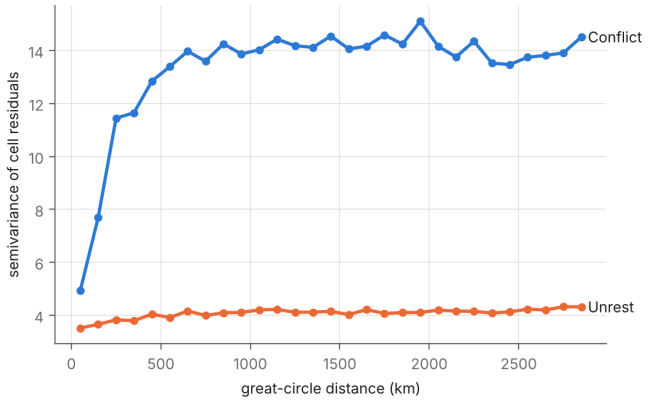


FIG S1. *Empirical semivariograms of the standardised test-period cell residuals, against great-circle distance.*

TABLE S5

S4: estimation of biome-specific temperature effects (30 replicates). "Detects sign": proportion of replicates in which the interval excludes zero on the correct side. ‡Median SHAP value of raw temperature within the biome, with the interquartile range of the bootstrap replicates, as in the original analysis; it estimates neither γ_b nor $\bar{\beta}_b$.

Estimator / biome	Mean est.	Cov. γ_b	Cov. $\bar{\beta}_b$	Detects sign
Biome slopes, no latent effects				
biome 1 ($\gamma = -0.3$)	-0.273	0.57	0.77	1.00
biome 2 ($\gamma = -0.1$)	-0.083	0.57	0.83	0.60
biome 3 ($\gamma = +0.1$)	0.108	0.53	0.70	0.73
biome 4 ($\gamma = +0.3$)	0.307	0.50	0.53	1.00
Biome slopes + u + w				
biome 1 ($\gamma = -0.3$)	-0.286	0.53	0.87	1.00
biome 2 ($\gamma = -0.1$)	-0.082	0.57	0.90	0.67
biome 3 ($\gamma = +0.1$)	0.119	0.60	0.73	0.73
biome 4 ($\gamma = +0.3$)	0.319	0.53	0.87	1.00
SVC: biome means + η + u + w				
biome 1 ($\gamma = -0.3$)	-0.277	0.97	1.00	0.93
biome 2 ($\gamma = -0.1$)	-0.077	0.93	1.00	0.23
biome 3 ($\gamma = +0.1$)	0.111	0.90	1.00	0.33
biome 4 ($\gamma = +0.3$)	0.309	0.83	1.00	0.97
XGBoost SHAP slope (targets $\gamma_b/2$)				
biome 1 ($\gamma = -0.3$)	-0.092	0.03	0.03	0.60
biome 2 ($\gamma = -0.1$)	-0.026	0.47	0.53	0.23
biome 3 ($\gamma = +0.1$)	0.048	0.47	0.40	0.37
biome 4 ($\gamma = +0.3$)	0.091	0.03	0.00	0.60
XGBoost SHAP slope, interactions credited				
biome 1 ($\gamma = -0.3$)	-0.191	0.27	0.40	0.83
biome 2 ($\gamma = -0.1$)	-0.056	0.63	0.63	0.30
biome 3 ($\gamma = +0.1$)	0.089	0.57	0.73	0.53
biome 4 ($\gamma = +0.3$)	0.176	0.40	0.30	0.80
XGBoost median SHAP, raw temperature‡				
biome 1 ($\gamma = -0.3$)	-0.052	0.03	0.07	0.57
biome 2 ($\gamma = -0.1$)	-0.151	0.13	0.13	0.97
biome 3 ($\gamma = +0.1$)	-0.150	0.00	0.00	0.00
biome 4 ($\gamma = +0.3$)	-0.043	0.00	0.00	0.20

TABLE S6

S4: recovery of the spatially varying coefficient $\beta(s)$ (mean and SD over 30 replicates).

Quantity	Mean	SD
$\mathrm{cor}\{\hat{\beta}(s), \beta(s)\}$, SVC	0.918	0.036
RMSE of $\hat{\beta}(s)$, SVC	0.106	0.010
RMSE of $\hat{\beta}(s)$, biome slopes only	0.147	0.020
$\hat{\sigma}_\eta$ [truth 0.15]	0.133	0.025
$\hat{\rho}_\eta$ (km) [truth 600]	564	261

TABLE S7

Test-period (2023–2024) residual diagnostics, Africa and the Middle East, for every model on identical test rows: Moran's I of the standardised cell residuals (permutation p-value from 999 permutations, bounded below by 0.001) and the residual correlation between neighbouring cells in the same country and across a border.

Outcome	Model	Moran's I	p	Neighbour residual corr.	
				within country	across border
Conflict	XGBoost, corrected	0.400	≤0.001	0.419	0.249
Conflict	XGBoost + cell event rate	0.336	≤0.001	0.344	0.247
Conflict	XGBoost, cross-fitted by country	0.554	≤0.001	0.601	0.246
Conflict	Cross-fitted XGBoost + u + iid cell effects	0.416	≤0.001	0.443	0.158
Conflict	Cross-fitted XGBoost + $u + w$	0.413	≤0.001	0.436	0.166
Unrest	XGBoost, corrected	0.138	≤0.001	0.150	0.090
Unrest	XGBoost + cell event rate	0.262	≤0.001	0.287	0.115
Unrest	XGBoost, cross-fitted by country	0.245	≤0.001	0.275	0.099
Unrest	Cross-fitted XGBoost + u + iid cell effects	0.246	≤0.001	0.270	0.086
Unrest	Cross-fitted XGBoost + $u + w$	0.239	≤0.001	0.262	0.080

TABLE S8

Fitted hyperparameters of the hybrid models, Africa and the Middle East: range ρ_w and standard deviation σ_w of the field, standard deviation of the independent cell effects, and standard deviation σ_u of the country effects.

Outcome	Latent terms	$\hat{\rho}_w$ (km)	$\hat{\sigma}_w$	$\hat{\sigma}_{\text{cell}}$	$\hat{\sigma}_u$
Conflict	$u + w$	264	1.47	–	0.80
Conflict	u + iid cell effects	–	–	1.30	1.40
Unrest	$u + w$	68	1.20	–	1.05
Unrest	u + iid cell effects	–	–	1.11	1.10

TABLE S9

Posterior summaries of the field w of the hybrid model averaged over regions: mean, standard deviation and posterior probability that the regional average is positive. The regions were chosen before the fit, as known conflict zones and regions expected to be quiet. The last two rows give the share of all cells whose posterior probability of $w(s) > 0$ is above 0.95 or below 0.05.

Region	Longitude	Latitude	Cells	Conflict		Unrest	
				mean (SD)	$P(>0)$	mean (SD)	$P(>0)$
Eastern DRC	27 to 30.5	-5 to 2	94	1.89 (0.19)	1.00	0.66 (0.11)	1.00
Northern Mozambique	38 to 41	-14 to -10	39	1.48 (0.30)	1.00	0.15 (0.20)	0.77
Syria and Iraq	36 to 46	32 to 38	240	2.52 (0.21)	1.00	-0.03 (0.10)	0.39
Yemen	43 to 49	13 to 17	84	1.85 (0.37)	1.00	0.12 (0.16)	0.75
Lake Chad	12 to 16	10 to 14	64	1.61 (0.19)	1.00	-0.23 (0.13)	0.05
Central Sahel	-5 to 5	11 to 17	240	1.46 (0.19)	1.00	-0.01 (0.09)	0.43
Horn of Africa	38 to 48	3 to 10	277	0.27 (0.21)	0.90	-0.04 (0.09)	0.33
Sudan	23 to 35	10 to 16	288	-0.06 (0.23)	0.41	0.08 (0.09)	0.82
Arabian interior	42 to 55	18 to 27	423	-0.60 (0.30)	0.02	-0.03 (0.09)	0.36
Western Congo basin	14 to 24	-5 to 3	320	-0.99 (0.22)	0.00	-0.18 (0.08)	0.01
Southern Africa	18 to 32	-34 to -22	619	-0.58 (0.25)	0.01	-0.00 (0.07)	0.48
Share of cells with $P\{w(s) > 0\} > 0.95$			13719	0.188		0.116	
Share of cells with $P\{w(s) > 0\} < 0.05$			13719	0.035		0.022	